

SOFTWARE ENGINEERING

Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #54

Local Sports Tournament & Bracket Manager

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Semester: 5

Section: I

USE-CASE FLOW SPECIFICATION

Use Case ID: UC-03

Use Case Name: Record Match Score

Primary Actor: Tournament Director

Supporting Actors: None

Goal: Record a valid final score, advance the correct team, and update tournament standings.

Preconditions: The Tournament Director is authenticated; the tournament and bracket exist; the selected match is scheduled and has not already been finalized.

Postconditions: The confirmed score is stored, the correct team advances, standings are recalculated, and the action is recorded in the audit log.

Main Success Scenario

1. The Tournament Director selects a scheduled match whose result has not yet been finalized.
2. The system displays the two participating teams and the score-entry fields.
3. The Tournament Director enters the final score and selects Confirm Result.
4. The system validates the score format and verifies that the match has not already been finalized.
5. The system stores the confirmed score and identifies the winner.
6. The system advances the correct team to the next bracket slot.
7. The system recalculates wins, losses, points, goal difference, and tournament standings.
8. If teams are tied, the system resolves the tie using the configured tiebreaker rule.
9. The system records the Tournament Director, action, and timestamp in the audit log.
10. The system displays a success confirmation with the updated bracket and standings. The updated information becomes available to Team Captains and Public Viewers after publication.

Alternate Flow - Invalid or Duplicate Score Submission

- 4a. If the score format is invalid, the system rejects the submission and displays a validation message. The Tournament Director corrects the score and returns to Step 3.
- 4b. If the match has already been finalized, the system rejects the duplicate submission and displays that the result is already recorded. No bracket or standings data is changed.
- 6a. If the winner cannot be placed in a valid next-round bracket slot, the system does not finalize the update and reports the bracket inconsistency to the Tournament Director.